

Writing Programs that are Creative

Procedural Programming Exercises

The first program ever written that was an attempt to make a machine behave creatively was a Love Letter Writing Program written by Christopher Strachey in 1952 (Strachey is the key person behind the way modern programming languages like Java are designed). You can see his program in action here: <http://www.gingerbeardman.com/loveletter/>

You can read about it at <https://cs4fn.blog/2022/06/20/the-algorithm-that-could-not-speak-its-name/>

It used templates and word lists to generate random love letters like

HONEY SWEETHEART,
MY AMBITION AFFECTIONATELY TREASURES YOUR DESIRE. YOU ARE MY PRECIOUS
SYMPATHY, MY SWEET AFFECTION. MY COVETOUS BURNING PANTS FOR YOUR
SEDUCTIVE ADORATION. YOU ARE MY TENDER HEART.
YOURS PASSIONATELY,
M.U.C.

DEAREST DEAR,
MY SEDUCTIVE ENTHUSIASM BEAUTIFULLY SIGHS FOR YOUR LOVE. YOU ARE MY
TENDER FONDNESS. MY FERVOUR YEARNS FOR YOUR BURNING WISH. MY AMOROUS
FONDNESS FONDLY CLINGS TO YOUR AFFECTIONATE HUNGER. MY FANCY HUNGERS
FOR YOUR EAGER LITTLE LIKING.
YOURS FONDLY,
M.U.C.

M.U.C here stands for the Manchester University Computer who wrote the love letters. It uses several different templates that it chooses randomly between like:

YOU ARE MY <adjective> <noun>.

1. Write your own love letter generating program that uses the same templates for sentences and word lists as Strachey's. Write it in a way that makes it easy to edit the word lists and sentence templates.
2. Now write your own love letter generating program that is **better** than Strachey's, writing more natural love letters. It can use different words in the word lists, more word lists, more or different sentence templates.
3. Choose a different kind of letter (eg Birthday card or Christmas card greetings) and write a generator for those.

My own attempt (so far) creates love letters like:

Dearest Sweetheart,
I want to caress you forever.
You are my sympathetic charm, my loving devotion.
I want to see you forever.
You are my peaches and cream.
I pine for your unsatisfied lust.
Yours, keenly
Q

Can you do better? You need to be good at both language and programming! Read more about natural language processing here <https://cs4fn.blog/natural-language-processing/> and creative computing here <http://www.cs4fn.org/creativity/>